Algorithm Types

Ten machine learning methods compared by learning paradigm, model form, application domain, and six relationships.

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Algorithm taxonomy

Supervised

The training examples include the outcome.

SUPERVISED

Decision Tree

Classical algorithm

● Tab · CV · NLP · GenAI

Splits labeled examples with feature tests and predicts at each leaf.

Use: Trace a small-loan decision through readable rules.

A

SUPERVISED

Random Forest

Ensemble method

● Tab · CV · NLP · GenAI

Averages many decision trees, each trained on a bootstrap sample of labeled data.

Use: Estimate readmission risk from clinical record fields.

A

SUPERVISED

Gradient Boosting

Ensemble method

● Tab · CV · NLP · GenAI

Adds trees one at a time, each fit to the remaining error of the ensemble.

Use: Estimate claim severity from policy and driver data.

A

SUPERVISED

Naive Bayes

Classical algorithm

○ Tab · CV · ● NLP · GenAI

Classifies by multiplying per-feature probabilities, assuming features are independent given the class.

Use: Filter spam from word-presence counts in labeled mail.

C

SUPERVISED

CNN

Neural architecture · usually trained with labels

Tab · ● CV · NLP · ○ GenAI

Reuses local filters across an image to build visual features.

Use: Flag chest X-rays that may need radiologist review.

F

Unsupervised

No externally supplied prediction label.

UNSUPERVISED

K-Means

Classical algorithm

● Tab · ○ CV · NLP · GenAI

Partitions observations into K groups around moving centroids.

Use: Segment customers by recency, frequency, and spend.

B

UNSUPERVISED

PCA

Classical algorithm

● Tab · ○ CV · NLP · GenAI

Rotates centered features into uncorrelated directions of high variance.

Use: Compress correlated sensor readings before plotting them.

B

Self-supervised

Inside unsupervised.

SELF-SUPERVISED

BERT

Pretrained model family · then supervised fine-tuning

Tab · CV · ● NLP · GenAI

Builds bidirectional token context, then fine-tunes on labeled tasks.

Use: Route support tickets from the words in context.

C D

SELF-SUPERVISED

GPT-style models

Pretrained model family · fine-tuning optional

Tab · CV · ● NLP · ● GenAI

Predicts the next token from earlier tokens, then continues the sequence.

Use: Draft a lab abstract one token at a time.

D E

SELF-SUPERVISED

Diffusion models

Generative model family · can be text-conditioned

Tab · ● CV · NLP · ● GenAI

Learns to reverse added noise over many denoising steps.

Use: Propose a product photo from a written brief.

E F

Supervised · Unsupervised · Self-supervised, drawn inside unsupervised · Square corners: classical algorithm or ensemble · Rounded corner: neural architecture or model family · ● primary domain, ○ secondary domain

Letter A–F: relationship, listed with the comparisons

Algorithm Types

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Relationships

A **Base learner: parallel vs. sequential**

Decision Tree → Random Forest / Gradient Boosting

Independent trees, then average.	Sequential trees fit the remaining error.

Both ensembles use trees. Forests train them separately. Boosting adds them to correct the current ensemble.

B **Groups rows vs. transforms columns**

K-Means ↔ PCA

Assigns each row to a cluster.	Replaces columns with variance axes.

Both skip prediction labels. Clustering does not reduce the feature count. PCA does not assign groups. They are often used together.

C **Independence vs. context (text)**

Naive Bayes ↔ BERT

Multiplies feature probabilities.	Reads each token from both sides.

Both can classify text. Naive Bayes is trained on labels. BERT is pretrained, then often fine-tuned.

D **Encoder vs. decoder (shared Transformer)**

BERT ↔ GPT-style models

Bidirectional masked-token encoder.	Causal next-token decoder.

Both are Transformers (Vaswani et al., 2017). Direction of context and the pretraining target differ.

E **Token-by-token vs. iterative denoising**

GPT-style models ↔ Diffusion models

Appends one token to a prefix.	Updates a whole sample from noise.

Text is a common GPT-style case and images a common diffusion case. Neither mechanism is only one modality.

F **Convolutional U-Net = denoiser backbone**

CNN → Diffusion models

Local filters learn visual features.	A U-Net predicts the added noise.

Ho et al. (2020) use a convolutional U-Net to denoise. Other diffusion backbones exist.

Classification matrix

Algorithm	Paradigm	Tabular Data	Computer Vision	Natural Language Processing	Generative AI
Supervised					
Decision Tree	Supervised	●	—	—	—
Random Forest	Supervised	●	—	—	—
Gradient Boosting	Supervised	●	—	—	—
Naive Bayes	Supervised	○	—	●	—
Convolutional Neural Network	Supervised	—	●	—	○
Unsupervised					
K-Means Clustering	Unsupervised	●	○	—	—
Principal Component Analysis	Unsupervised	●	○	—	—
Self-supervised · inside unsupervised					
BERT	Self-supervised	—	—	●	—
GPT-style Autoregressive Models	Self-supervised	—	—	●	●
Diffusion Models	Self-supervised	—	●	—	●

Supervised · Unsupervised · Self-supervised, drawn inside unsupervised · Square corners: classical algorithm or ensemble · Rounded corner: neural architecture or model family · ● primary domain, ○ secondary domain · Letter A–F: relationship, listed with the comparisons

Self-supervised: training targets are constructed from unlabeled data (masked token, next token, added noise); classified as unsupervised for this assignment. Generative AI is a task category. Use cases are illustrative. Mechanism diagrams and full references: https://sriharshitha-ayyalasomayajula.github.io/iwu-academic-portfolio/